Thin-Wall Turn Around Manifold Directed Energy Deposition Component
Statement of Work (SOW)

NASA's Marshall Space Flight Center (MSFC) requires a large-scale manufacturing technology demonstration using thin-wall Laser Powder Directed Energy Deposition (DED). This effort aims to advance the partial developmental fabrication of a JBK-75 RS-25 thin-wall turnaround manifold, integral to the aft-end nozzle.

The SOW is as follows:

Task 1

1. The vendor shall fabricate and provide engineering support for the laser powder directed energy deposition (LP-DED) of an RS-25 aft nozzle turnaround manifold with an approximate wall thickness of 1 mm.
2. The vendor shall prepare a large-scale build plate support an aft end of a thin-wall RS-25 nozzle demonstrator.
3. The vendor shall assess machine setup and build plate weight factors affecting the nozzle build and provide recommendations for toolpaths.
4. The vendor shall review final models and refine the build approach for the JBK-75 thin-wall RS-25 nozzle demonstrator, focusing on the aft end.
5. The vendor shall provide design inputs and develop detailed toolpath strategies for confirmation by NASA.
6. Vendor shall demonstrate LP-DED using a large scale machine platform for capability to print the aft end of a RS-25 nozzle with a combined height of greater than 90” height and greater than 80” diameter.
7. Vendor shall provide capability within laser system for 1 mm wall thickness at diameters stated in Task 1, requirement #6.
8. The material shall be JBK-75 Fe-Ni alloy per NASA requirements. The use of virgin powder, 1x used and 2x used is acceptable.
9. The print shall be performed within 55 consecutive days, or approximately 1,000 hours of machine time shall include the following:
    a. Engineering Support for a dedicated 557 laser system (includes Tool Code generation, part design input, laser scan inspections, reporting, etc.)
    b. Provide operator for machine setup and deposition.
    c. Provide consumables, such as Argon nozzles, filters, and all utilities required for operation.
    d. Provide a build volume at a minimum of 60” dia by 72” for component builds.
    e. Provide an inert gas chamber as part of the build to minimize oxidation. Provide all argon and consumables.
    f. Provide tool code required to complete large format samples.
    g. Provide all maintenance on the machine during the period of builds.
    h. Provide operators necessary to operate the machine.
    i. Shall include other required direct costs such as consumables and necessary

replacement components (inert gas, build-nozzle components, filters, etc).
   j. All engineering support should be included as part of the continuous-build machine capacity. This includes all tool path generation, inspections, trouble shooting of build failures, reporting, etc.
   k. Shall include removal of parts from the build plate as required.
10. Vendor shall complete a minimum deposition of 70% of the thin-wall large scale RS-25 turn around aft nozzle.
11. Part should be run on the same machine consecutively with acceptable breaks as needed for expected or unexpected shutdowns or pauses.
12. NASA shall provide drawings, models, and dimensions as required.
13. NASA will provide all heat treatments and characterization and inspections as necessary.
14. NASA shall provide DED-sized powder, specifically JBK-75, for the builds.

Shipping

1. Contractor shall ship all deliverables and process development samples to:
   Marshall Space Flight Center
   Central Receiving, Bldg 4631
   [redacted]
   Huntsville, AL 35812

Period of Performance (POP)

- 6 months from authority to proceed (ATP)

Deliverables

1. DED samples and any reporting from Task 1 on the large-scale aft end turn around manifold build.